

Michael Fischbach

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OBJECTIVE

Mechanical Engineering student (Cal Poly Honors, 3.78 GPA) seeking the **Praktikum im Bereich Gesamtfahrzeugentwicklung at HWA AG** (Affalterbach, from Feb/March 2027). Hands-on CAD-to-part experience across 4-axis CNC machining, prototyping, tolerance analysis, and iterative real-world testing, the same design-workshop-testing loop HWA runs on the EVO and its customer racing programs. Eligible to intern in Germany.

EDUCATION

California Polytechnic State University, San Luis Obispo

Sep 2025 – Jun 2029

B.S. Mechanical Engineering, Honors Program | GPA: 3.78 | Dean's List: Fall, Winter, Spring

- Honors research-track program; Learn by Doing curriculum with hands-on manufacturing, CNC machining, and design from first year
- Built multi-component mechanisms (piston, Geneva wheel, CNC-machined screwdriver) with full tolerance stack-up analysis and GD&T

Oaks Christian School – Institute of Engineering

Sep 2022 – Jun 2025

Engineering Pathway: 6 multi-month applied design projects | GPA: 3.92 UW / 4.10 W

TECHNICAL SKILLS

CAD / CAE: Fusion 360 (6+ years, 40+ designs), SolidWorks, AutoCAD, Autodesk CFD

Manufacturing: 4-axis HAAS CNC machining, Bridgeport mill, lathe, FDM/SLA 3D printing, carbon-analog composite & concrete casting

Methods: GD&T, tolerance stack-up analysis, DFM/DFA, rapid iterative prototyping, design-test-refine, cost modeling

Software: Python, HTML/CSS, Git

Languages: English (native); German (beginner, actively learning)

MANUFACTURING & PROTOTYPING PROJECTS

CNC Milled Golf Putter – Full CAD-to-Part Build

2024 – 2025

4-Axis HAAS CNC, Fusion 360, DFM

- Designed and machined a novel putter geometry on a 4-axis HAAS CNC; ran iterative PLA prototypes to validate geometry before the final aluminum production part
- Performed DFM and cost analysis: modeled warehouse-scale unit cost at \$15.78 vs. \$50–150 retail (533% reduction)
- Selected 1 of 8 finalists from 40+ students; presented the build and process at Capstone Night to 200+ alumni and industry professionals and several hundred students

Wave Energy Converter – Rack & Pinion Generator

2023

Bridgeport Mill, Fusion 360, Fluid Dynamics

- Designed and machined a wave-oscillation-to-electrical-power generator with a five-person team; manufactured all components to tolerance on a Bridgeport mill
- Took 1st place with the highest measured output of three teams: 15 V per motor at 1:12 scale, 360 V scaled
- Applied fluid dynamics and energy-transfer analysis; modeled the full assembly in Fusion 360 and AutoCAD with tolerance documentation

Chameleon Ramps – Product Design & Manufacturing

2019 – Present

Fusion 360, Casting, Iterative Prototyping

- 6-year iterative development program: designed 40+ unique products and developed proprietary casting techniques, reducing material waste ~30%

- Ran a systematic design-test-refine loop across 4–6 prototype generations per product; generated \$50,000+ in revenue

Autonomous Execution System
2025 – Present

Python, Real-Time Data, SQLite, Backtesting

- Built an autonomous Python system that ingests real-time data feeds and executes decisions through an external API, with risk limits, state persistence, and a custom backtesting harness
- **1553 tests passing** in production; parameters selected on their worst case across two disjoint validation windows rather than their best

EXPERIENCE

CEO & Founder
Jun 2026 – Present

Campus Native
San Luis Obispo, CA

- Founded and operate a digital marketplace; lead a 4-person intern team across engineering, marketing, and content

Digital Media Manager
Mar 2023 – Present

All Domain Integration (ADI) – Aerospace Engineering
Remote

- 3+ year engagement with an aerospace engineering firm: website management, SEO strategy, and digital communications

Business Partner
May 2023 – Jul 2024

Ashersells LLC
Malibu, CA

- Helped scale operations to \$500,000+ annual revenue; built Python automation for logistics and order processing

LEADERSHIP & INTERNATIONAL PROFILE

- **Cal Poly Honors Program** – Admitted Fall 2026; selective research-focused program with faculty mentorship
- **Eagle Scout** – Boy Scouts of America (2025); Eagle rank, Senior Patrol Leader 2 years; 15 years leadership
- **Dean's List** – Cal Poly SLO, all three quarters (Fall / Winter / Spring 2025–26)
- **Internationally Mobile** – U.S. Air Force family; 9 relocations across 7 states and territories; adapts quickly to new environments and teams
- **Community Service** – 4+ years bimonthly meal service to unhoused community; 2021–present